



TRAINING PROGRAMME

Data Engineering with Spark & Kafka

Batch and streaming pipelines with Apache Spark, Apache Kafka and lakehouse tables

33 HOURS

FOR Data analysts, developers and engineers moving into data engineering



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Overview

This programme teaches participants to build batch and streaming data pipelines with open-source tools and Confluent Schema Registry. They see how big data processing moved from HDFS and MapReduce to object storage and the lakehouse, process data with Spark DataFrames and Spark SQL, store it in Delta Lake and Apache Iceberg tables, and tune Spark jobs. They then run Kafka on KRaft, process event streams with Spark Structured Streaming, integrate systems with Kafka Connect and Schema Registry, and schedule and check pipelines with Airflow and data quality tests. Every chapter ends with a lab on a local stack, and the programme closes with a final project: real-time order analytics built and presented by each team.

At a glance

#	PHASE	FOCUS	HOURS
01	Spark Foundations	From Hadoop to the lakehouse, Spark architecture, DataFrames and Spark SQL	6 h
02	Lakehouse Tables & Performance	Parquet, Delta Lake and Iceberg, partitioning, joins, adaptive query execution	6 h
03	Streaming, Integration & Orchestration	Kafka on KRaft, Structured Streaming, Kafka Connect, Schema Registry, Airflow	9 h
04	Final Project	Real-time order analytics, presented to the group	6 h
	Buffer	Catch-up, revision, extra lab time	6 h
Total			33 h

Key facts

AUDIENCE	Data analysts, developers and engineers moving into data engineering
PREREQUISITES	Python, pandas and SQL (course: Python for Data Science), or equivalent. These are used, not taught
DURATION	33 h: 27 h of chapters + 6 h buffer
FORMAT	Short lectures followed by guided labs on one dataset carried through every phase
LAB ENVIRONMENT	Runs locally: one laptop per participant (16 GB RAM minimum, Docker, VS Code, Jupyter) with a Docker Compose stack prepared by the trainer: Spark, Kafka 4 (KRaft), Schema Registry, Kafka Connect, Airflow 3, MinIO and an Iceberg REST catalog, services started per phase; no cloud account or cost
ASSESSMENT	A lab validated at the end of each phase and a final project presented to the group; certificate of completion for participants who complete both
OUTCOME	Participants can build, tune and schedule batch and streaming pipelines with Spark and Kafka that write reliable lakehouse tables and pass data quality checks

PHASE 01

6 H

Spark Foundations

Participants see why data platforms moved from HDFS and MapReduce to object storage and the lakehouse, learn how Spark plans and runs work across a cluster, then transform data with DataFrames, Spark SQL and user-defined functions. Validated lab: a Spark job that cleans and aggregates the course dataset from object storage.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	Big Data Foundations	From HDFS and MapReduce to object storage and the lakehouse	Lecture + Workshop	1.5 h
1.2	Spark Architecture	Driver, executors, jobs, stages, lazy evaluation	Lecture + Lab	1.5 h
1.3	DataFrames & Spark SQL	DataFrame API, Spark SQL, user-defined functions	Lecture + Lab	3 h

PHASE 02

6 H

Lakehouse Tables & Performance

Participants store results in open table formats that bring transactions, time travel and schema evolution to object storage, then make Spark jobs faster with partitioning, caching, the right joins and adaptive query execution, reading the Spark UI to find the bottleneck. Validated lab: a Delta or Iceberg table updated with a merge and read at an earlier version, and a slow job made measurably faster.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Table Formats	Parquet, Delta Lake and Apache Iceberg: ACID tables, time travel, schema evolution	Lecture + Lab	3 h
2.2	Spark Performance	Partitioning, caching, joins, adaptive query execution, Spark UI	Lecture + Lab	3 h

PHASE 03

9 H

Streaming, Integration & Orchestration

Participants run Kafka on KRaft and learn topics, partitions, producers and consumer groups, process event streams with Spark Structured Streaming using windows and watermarks, connect external systems with Kafka Connect and enforce schemas with Schema Registry, then schedule pipelines with Airflow and add data quality checks. Validated lab: a stream of events aggregated per minute into a lakehouse table, with late events handled by a watermark.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Kafka Fundamentals	KRaft clusters, topics, partitions, producers, consumer groups, delivery semantics	Lecture + Lab	3 h
3.2	Structured Streaming with Kafka	Streaming queries, windows, watermarks, sinks, exactly-once output	Lecture + Lab	3 h
3.3	Kafka Connect & Schema Registry	Connectors, schemas and compatibility	Lecture + Lab	1.5 h
3.4	Orchestration & Data Quality	Airflow DAGs, data quality checks	Lecture + Lab	1.5 h

PHASE 04

6 H

Final Project

Teams of two or three build a real-time analytics pipeline for an online shop's orders, reusing the phase labs, then present it as they would to a data team lead.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Brief & Build	Read the brief, produce order events to Kafka with a registered schema, aggregate them with Structured Streaming into Delta or Iceberg tables	Project	3 h
4.2	Schedule, Check & Document	A daily batch job scheduled by Airflow, data quality checks, a Jupyter notebook dashboard and a short design note	Project	1.5 h
4.3	Demo & Review	Live demo with a burst of orders; questions and feedback	Project	1.5 h

FINAL PROJECT

Real-Time Order Analytics

Order events produced to Kafka with a registered schema, aggregated by Spark Structured Streaming into Delta or Iceberg tables, completed by a daily batch job scheduled in Airflow, checked by data quality tests and shown on a dashboard.

STACK Apache Kafka (KRaft), Schema Registry, Apache Spark (Structured Streaming, Spark SQL), Delta Lake or Apache Iceberg, MinIO, Apache Airflow, Python, Jupyter

DELIVERABLE Pipeline code, design note, dashboard, live demo